Education and Poverty Across U.S. Counties

ECON 3916 Midterm Project

Section 1: The Question and the Data

Question

Do U.S. counties with a higher share of adults (25+) with a bachelor's degree have lower poverty rates after controlling for rural–urban status, median age, and state?

Why it matters, and to whom

The state and county invest heavily in higher education and workforce programs because they expect them to reduce educational poverty. If the poverty rate remains high despite having more college graduates, this could change how local economic development offices, state legislators, and poverty alleviation agencies use their budgets. This question is also very important in rural areas with low educational attainment and high poverty rates. It shows whether education explains part of this gap, or whether rural residence is also important.

Data sources

ACS (S1701, S1501, B01002): The U.S. Census Bureau collects this data through the American Community Survey (ACS). I use a five-year estimate from 2020 to 2024. ACS provides information on communities between decennial censuses and supports federal funding allocation. Each county figure is a survey estimate based on five years of data and is not an aggregate.

RUCC 2023: The USDA Economic Research Service classifies counties into nine codes based on metropolitan size, urban population, and proximity to a metropolitan area. Codes 1–3

represent metropolitan counties, and codes 4–9 represent non-metropolitan counties. This classification uses the 2020 census and the 2023 OMB definition of a metropolitan area.

I used five-digit county FIPS codes to combine three ACS tables with the RUCC files. After excluding 78 Puerto Rican municipalities, 3,144 counties matched. The data was downloaded on November 3, 2026, from data.census.gov and ers.usda.gov.

The data as a product of measurement decisions

Each variable reflects a specific measurement selection.

- The poverty rate uses a federal poverty line that applies the same dollar standard nationwide. Differences in living costs between areas such as rural Mississippi and San Francisco are not reflected.
- The bachelor's ratio includes only adults aged 25 and over. This measures where graduates currently reside, not where they grew up or studied.
- All three ACS variables are five-year survey estimates for 2020–2024 and are not aggregate data. Each indicator has a margin of error (MOE), and the range tends to be larger in smaller counties. While the mean MOE for the county poverty rate is ±2.6, the MOE for the proportion of bachelor’s degree holders in Loving County is ±66.1.
- RUCC is a USDA classification based on commuting patterns and urban population. Since the code is categorical, I use dummy variables instead of treating it on a scale from 1 to 9.

Unit and weighting

Each row represents a county, and there are 3,144 counties in total, including the 50 states and Washington, D.C. The main model assigns equal weights to all counties. In other words, Loving County (33 people) has the same population as Los Angeles County (9.7 million). Since I am using a county as my research subject, this aligns with my research question. Chapter 3 also introduces a population weighting model as supplementary material. That model focuses on the average person.

Missingness

There are no missing values in the final sample. Puerto Rico (78 municipios) and 13 RUCC records for other U.S. territories (American Samoa, Guam, Northern Mariana Islands, and U.S. Virgin Islands) were excluded by design because my study only covers the 50 states and DC.

However, missing responses within the ACS may still affect the results. The Census Bureau uses weighting and imputation to account for households that do not respond. This approach assumes that nonresponse is missing at random (MAR) after accounting for factors such as location and household characteristics. However, some nonresponse may be missing not at random (MNAR). If the poorest households are still less likely to respond, even within similar groups, poverty rates may be underestimated, especially in small and poor counties.

What this does to my answer

If poverty is undercounted in small low-income rural counties due to lack of response, the gap between low- and highly educated counties may appear smaller than it actually is. This will bring the coefficient for the percentage of bachelor's degree holders to near zero.

Measurement errors in the proportion can also affect the results. Smaller counties tend to have less accurate estimates; for example, Loving County's MOE is large at ±66.1 points. If this measurement error is random and unrelated to the actual proportion of bachelor's degrees or

the error term, the estimated coefficient can be biased toward zero. In other words, the estimated negative relationship may be smaller than the actual relationship. Therefore, my estimate may be somewhat conservative.

What the summary table shows

Table 1. Summary statistics, U.S. counties

	N	Mean	SD	Min	Median	Max	Pop.–weighted mean	
Poverty rate (%)	3144.0	14.2	5.9	1.4	13.2	57.6	12.5	
Adults 25+ with a bachelor's degree or higher (%)	3144.0	24.7	10.3	0.0	22.2	80.5	35.6	
Median age (years)	3144.0	41.8	5.4	21.4	41.5	86.2	39.1	
Population	3144.0	104032.8	329658.0	33.0	25173.5	9656697.0	NaN	
Margin of error of poverty rate (± points)	3144.0	2.6	1.7	0.2	2.2	21.1	0.9	
Margin of error of bachelor's share (± points)	3144.0	2.5	1.9	0.2	2.2	66.1	1.0	

Source: ACS 2020–2024 5-year estimates. One row = one county (50 states + DC).
Margin of error = Census 90% margin. Pop.–weighted mean weights each county by its population.

The average county has a poverty rate of 14.2% (SD 5.9) and a bachelor's share of 24.7% (SD 10.3). When weighted by population, poverty falls to 12.5%, and the bachelor's share rises to 35.6%. This suggests that larger counties tend to have higher education levels and lower poverty rates.

Measurement is also more precise in larger counties. The average poverty-rate MOE is 2.6 points across counties but only 0.9 points when weighted by population. Table 2 (Appendix) also shows that 62% of counties (1,958) are nonmetro. Average poverty rises from 10.6% in the largest metro counties (code 1) to about 15% in nonmetro counties.

Section 2: Descriptive Evidence

Figure 1. Most counties have 10-17% poverty, but a few are far above

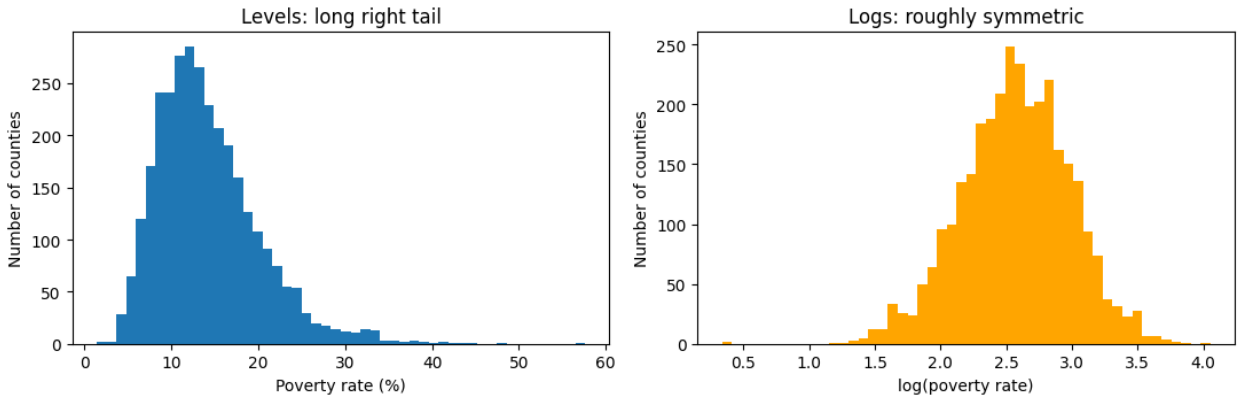


Figure 1. Poverty rates are right-skewed, but log poverty is nearly symmetric. Source: ACS 2020–2024 5-year, table S1701. N = 3,144 counties.

Half of the counties have poverty rates between 10% and 17%, while 18 counties exceed 35% (skew = 1.19). After taking the log, the distribution becomes nearly symmetric (skew = -0.21).

What I need to do: Keep poverty at different levels in the main model to make the coefficients easy to interpret as percentage points. The skewness of the outcome variable does not affect the OLS coefficients. Since the main concern is heteroscedasticity, I used the standard error clustered by week. In Chapter 3, I use log poverty as a robustness-verified indicator.

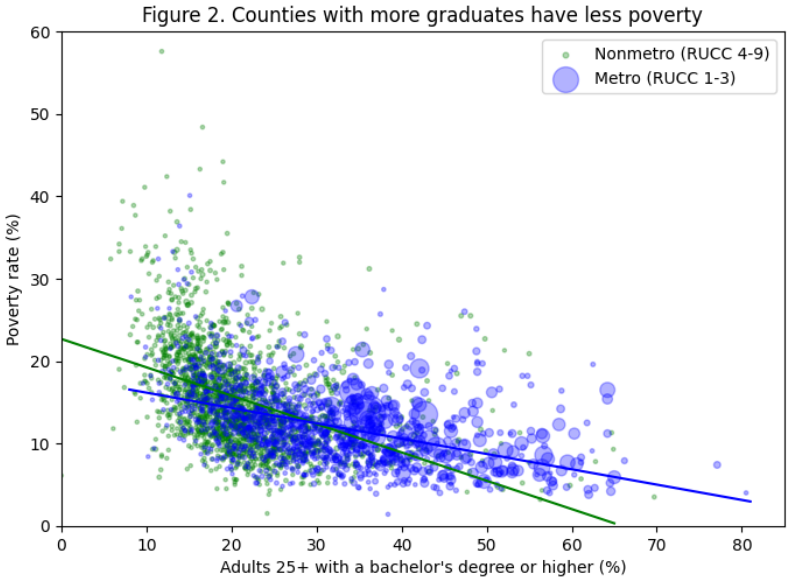


Figure 2. Counties with more graduates tend to have lower poverty, especially in nonmetro areas. Lines are OLS fits within each group. Dot size represents population. Source: ACS 2020–2024 5-year (S1701, S1501); USDA ERS RUCC 2023.

The raw correlation coefficient is -0.46. In nonmetro counties, a 1 percentage point increase in the percentage of bachelor's degree holders is associated with a roughly 0.34 percentage point decrease in the poverty rate. In metropolitan counties, the decline is about 0.19 percentage points. Many non-urban counties have lower levels of education and higher poverty rates. This is why I must control for the rural–urban state before interpreting this relationship as an educational effect.

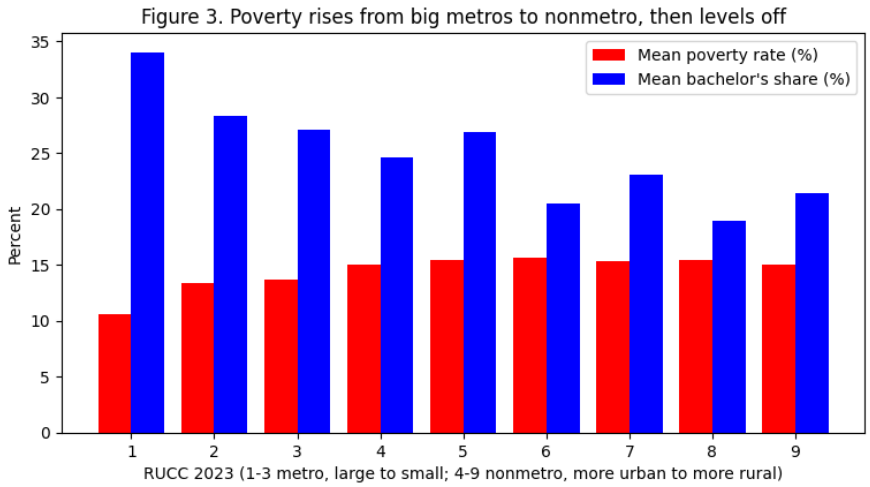


Figure 3. Poverty rises across metro and nonmetro counties, while education does not follow a clear pattern. Unweighted county means by 2023 Rural-Urban Continuum Code. Source: ACS 2020–2024 5-year; USDA ERS RUCC 2023.

In metropolitan counties with populations over one million, the average poverty rate rises from 10.6% to about 15% based on code 4. After that, it maintains nearly the same level through code 9. However, the proportion of those with a bachelor's degree does not steadily decline. For example, counties in the 5, 7, and 9 codes have higher levels of education than neighboring categories. Since neither pattern connects the RUCC codes in a straight line, I used eight dummy variables instead of treating the RUCC as a single scale from 1 to 9. This follows the feedback from checkpoint 1.

Section 3: The Model

The model

Table 3 (Appendix) shows the OLS results for county poverty rates. Column (1) includes only bachelor's share. Columns (2) to (4) add state dummies, RUCC dummies, and median age one by one. Column (4) is my main model. Standard errors are clustered by state.

Interpreting the bachelor's-share coefficient

Compare two counties that share the same state, rural-urban status, and median age. If the percentage of bachelor's degree holders in a county is 1 percentage point higher, the poverty rate in that region is on average 0.19 percentage points lower. The 95% confidence interval is 0.15 percentage points lower than 0.22 percentage points, indicating that this relationship is statistically significant. Counties with a bachelor's degree rate one standard deviation higher (10.3) are about 1.9 points lower in poverty. This is about 13% of the average county poverty rate of 14.2%. This control reduces the coefficient by about 30% at temperature (1), making it -0.27. This suggests that some of the underlying relationships in Figure 2 stem not only from education but also from state-level and rural–urban differences. However, this is merely an association, not causation. College graduates may already be relocating to places with better job opportunities.

Interpreting the median-age coefficient

Holding education level, rural-to-urban ratio, and state constant, counties with a higher median age have a poverty rate 0.29 percentage points lower (95% CI: 0.20–0.37 percentage points lower). Counties with 1 standard deviation (5.4 years) more are about 1.5 percentage points lower in poverty rate. One possible reason could be that the older the county, the more retirees receive Social Security benefits and remain above the poverty line. They also tend to have fewer children, and those children have the highest poverty rates.

Interpreting a RUCC coefficient

Counties in the most remote category (code 9) have poverty rates 4.8 percentage points higher than large metro counties (code 1), at the same education level and median age (SE = 0.62). This suggests that rural areas have higher poverty for reasons beyond differences in education.

Justifying each control

- State dummies: control for confounding factors. States have different policies and living costs that affect both education and poverty. Tuition policies and university systems affect education levels, while Medicaid expansion, minimum wages, and state EITCs affect poverty. Without state dummies, the education coefficient could reflect these state differences. State is not a bad control because education cannot change which state a county belongs to. I also cluster standard errors by state because counties in the same state share many policies.
- RUCC dummies: control for rural–urban differences. Rural areas tend to have fewer jobs that require a degree and offer lower wages. This can lead to both lower education levels and higher poverty. Without RUCC, the education coefficient may be too negative. I use eight dummy variables because RUCC codes are categories, not a numerical scale (Figure 3). However, RUCC could be a bad control if higher education helps a county grow and become more urban. In that case, rurality is partly a mediator between education and poverty. Controlling for it would remove part of the relationship. For this reason, column (7) of Table 4 excludes RUCC. The coefficient changes from -0.19 to -0.25. The first estimate compares counties with similar rurality, while the second also includes the part of the relationship related to urbanization.
- Median age: controls for age differences and improves precision. Retirement areas and older farming counties may have different education levels. Age also affects poverty because of Social Security and differences in the number of children. Adding median age increases R^2 from 0.425 to 0.476 between columns (3) and (4). This means age explains some additional variation in poverty. However, age could be a bad control if college graduates move into an area and lower its median age. The education coefficient barely changes after adding age (-0.182 to -0.186), so this does not seem to have a major effect on my results.

Robustness (Table 4)

The main model assigns the same weight to all counties. My question is focused on the county. In the fifth column, weights were applied to counties based on population to focus on the average person. The coefficient is -0.15; it is smaller but still negative and statistically significant. Even after excluding Loving County in the 6th row, the coefficient remained almost unchanged (-0.187). Finally, use log poverty as the outcome variable for the eighth column. The poverty rate was found to decrease by about 1.5 percentage points for every 1 percentage point increase in the percentage of people holding a bachelor's degree. This suggests that the main outcome is not affected by the distorted distribution of poverty rates shown in Figure 1.

Section 4: Diagnostics

Residual plots (Figure 4, Appendix)

Graphically, the residuals are centered at zero, but the distribution widens as predicted poverty increases. Counties predicted to have a poverty rate of about 10% show a small error, while those predicted to be 20% or higher show a much larger error. Regarding the bachelor's degree ratio, the residuals are also distributed around 0, showing no distinct curve. The largest lag occurs in low-education counties (10–20%). These counties have much higher poverty rates than the model predicted. Overall, the linear form seems valid, but the constant variance assumption does not hold.

Constant variance

The results of the Breusch-Pagan test are consistent with the observations made in the residual graph (LM = 435.9, $p < 0.001$). I reject the assumption that dispersion is constant. In other words, the standard error of the ordinary OLS may be too small. I use the standard error clustered by week. This takes unequal dispersion into account, enabling the interrelation of

errors between counties within the same state for analysis. The cluster standard error for the proportion of bachelor's degree holders is 0.017, which is higher than 0.010 for the standard error of ordinary OLS and 0.011 for the HC1 robust standard error. Aggregation reduces the accuracy of estimates, but it is a more cautious approach. Nevertheless, the 95% confidence interval (-0.22 to -0.15) is still far from zero.

Variance inflation factors (VIF)

The VIF of the proportionality is 1.68. In other words, the standard error is about 30% larger than if I had assumed that education was independent of the other variables ($\sqrt{1.68} \approx 1.3$). This is a relatively small increase. Because rural counties tend to have lower levels of education, the level of education overlaps to some extent with the control variable. However, the degree of overlap is not large enough to make the estimate unreliable. The mean VIF is 1.32, and the VIF of the largest RUCC pile is 2.96. All figures are below the typical warning thresholds of 5 or 10. Therefore, multicollinearity does not appear to be a major problem.

Influential observations

Using Cook's rule and the 4/N rule (cutoff = 0.0013), I selected 130 influential counties from a total of 3,144. The most influential areas are Ogallala and Todd counties in South Dakota, and Sioux County in North Dakota. These are reservation counties with poverty rates between 43% and 58%, far higher than model projections based on education level, rural status, and state characteristics. Kala Wao County, Hawaii (population 66), also has a high Cook's distance.

The reason DC is excluded from the Cook's distance ranking is that it is the only county-level observation in that state. The status dummy fits perfectly, making the leverage 1 and the residual value 0. Since the crosstrike is divided by (1 - leverage), the calculation for DC is not defined. The model was also re-estimated while keeping all coefficients, except for DC and the bachelor's degree ownership ratio coefficient, unchanged.

Even after excluding all 130 influential counties, the proportion of bachelor's degree holders only changed slightly from -0.186 to -0.177 (SE = 0.013). This suggests that the main outcome does not depend on a small number of unusual counties. However, these influential counties point out an important factor not included in the model: reservation status and its historical background. This topic is discussed in more detail in Chapter 5.

Difference-in-differences

My data are a single cross-section, with no before-and-after period or policy change. Therefore, a difference-in-differences design does not apply to my research question.

Section 5: Threats and Conclusion

For the bachelor's degree holding rate coefficient to have a causal meaning, it must remain unrelated to other factors affecting education-related poverty even after controlling for state, rural-urban type, and median age. However, this is unlikely to be true. This means that a lower poverty rate may in part contribute to higher education levels, as college graduates are more likely to move to counties with better job opportunities.

Other factors such as industrial composition, regional job demand, and each county's economic and racial history also affect education and poverty, but they were not included in the model. The influential counties discussed in Chapter 4, primarily reservation counties with poverty rates far above projections, reveal some factors the model may be missing. Estimates for small counties may be unreliable, and poverty in low-income areas could be underreported due to non-response from households. These issues may weaken the presumed relationship, but that does not mean it is causation.

Therefore, this coefficient should be interpreted as a conditional association rather than a causal relationship. Within the same state, among counties with the same rural-to-urban type and median age, a 1 percentage point increase in the proportion of bachelor's degree holders

was associated with an average 0.19 percentage point decrease in the poverty rate. This graph shows how education is associated with poverty in each U.S. county, but it does not indicate what changes would occur if the proportion of college graduates in a county were increased. In conclusion, higher education counties tend to have lower poverty rates even after controlling for state, rural-urban type, and median age. A 10 percentage point increase in the proportion of bachelor's degree holders is associated with a reduction in the poverty rate of about 1.9 percentage points. This relationship is fairly consistent across various model specifications. Excluding 130 influential counties (-0.18), Loving County, and even when using log values, the results remain significant at the state-level standard error (95% confidence interval: -0.22 to -0.15) and with population weighting (-0.15). Depending on whether ruralness is a disturbance or a mediator variable, the estimate ranges from -0.19 to -0.25. However, these results do not prove that education leads to a reduction in poverty.

With more time, industry composition and hold indicators will be included as additional control variables. I would also use data from multiple ACS periods to follow the same counties over time and account for characteristics that do not change much. Finally, I would look for outside factors that affected education levels but were not directly chosen by counties, such as the historical locations of public universities, to better understand whether education has a causal effect on poverty.

Appendix

Table 2. Counties by 2023 Rural-Urban Continuum Code

rucc	rucc_label	counties	mean_poverty	mean_ba_share
1	**Metro - Counties in metro areas of 1 million population or more**	443	10.6	34.0
2	**Metro - Counties in metro areas of 250,000 to 1 million population**	385	13.4	28.3
3	**Metro - Counties in metro areas of fewer than 250,000 population**	358	13.7	27.1
4	**Nonmetro - Urban population of 20,000 or more, adjacent to a metro area**	201	15.0	24.6
5	**Nonmetro - Urban population of 20,000 or more, not adjacent to a metro area**	76	15.4	26.9
6	**Nonmetro - Urban population of 5,000 to 20,000, adjacent to a metro area**	379	15.6	20.5
7	**Nonmetro - Urban population of 5,000 to 20,000, not adjacent to a metro area**	247	15.3	23.1
8	**Nonmetro - Urban population of fewer than 5,000, adjacent to a metro area**	466	15.5	18.9
9	**Nonmetro - Urban population of fewer than 5,000, not adjacent to a metro area**	589	15.1	21.4

	county	pop	poverty_rate	ba_share	ba_moe	median_age	rucc
205	Los Angeles County, California	9656697	13.7	36.0	0.2	38.2	1
2675	Loving County, Texas	33	6.1	0.0	66.1	86.2	9

Table 3. OLS estimates of the county poverty rate (%)

	(1)	(2)	(3)	(4) Main
ba_share	−0.267***	−0.226***	−0.182***	−0.186***
	(0.025)	(0.020)	(0.019)	(0.017)
median_age				−0.286***
				(0.043)
C(rucc)[T.2]			1.278***	1.409***
			(0.351)	(0.359)
C(rucc)[T.3]			2.097***	2.212***
			(0.396)	(0.355)
C(rucc)[T.4]			3.028***	3.144***
			(0.447)	(0.423)
C(rucc)[T.5]			3.966***	3.641***
			(0.554)	(0.589)
C(rucc)[T.6]			2.611***	3.095***
			(0.432)	(0.421)
C(rucc)[T.7]			3.349***	3.897***
			(0.569)	(0.538)
C(rucc)[T.8]			2.292***	3.611***
			(0.421)	(0.397)
C(rucc)[T.9]			3.251***	4.785***
			(0.575)	(0.617)
Intercept	20.818***	23.326***	20.398***	31.765***
	(0.958)	(0.399)	(0.525)	(1.953)
R-squared	0.213	0.401	0.425	0.476
R-squared Adj.	0.213	0.391	0.414	0.466
N	3144	3144	3144	3144
State dummies	No	Yes	Yes	Yes

Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01
Standard errors clustered by state in parentheses. * p<0.1, ** p<0.05, *** p<0.01
RUCC base category: code 1 (metro areas of 1 million or more). State dummies not shown.

Table 4. Robustness checks

	(4) Main	(5) Pop.-weighted	(6) No Loving	(7) No RUCC	(8) Log poverty
Bachelor's share (pp)	−0.186***	−0.152***	−0.187***	−0.249***	−0.015***
	(0.017)	(0.015)	(0.017)	(0.020)	(0.001)
Median age (years)	−0.286***	−0.479***	−0.285***	−0.224***	−0.018***
	(0.043)	(0.071)	(0.044)	(0.040)	(0.002)
RUCC dummies	Yes	Yes	Yes	No	Yes
N	3,144	3,144	3,143	3,144	3,144
R-squared	0.476	0.509	0.476	0.437	0.499

Dependent variable: poverty rate (%) in columns 4–7; log poverty rate in column 8. All models include state dummies. Standard errors clustered by state in parentheses. p<0.10, p<0.05, ** p<0.01.

Figure 4. The spread of the residuals grows with fitted poverty

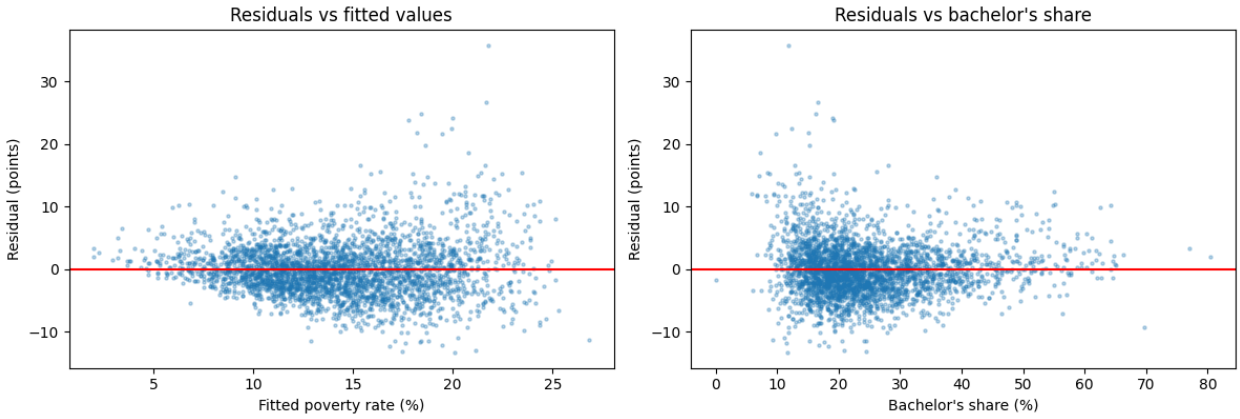


Figure 4. The spread of the residuals grows with fitted poverty. Residuals of the main model (4) against fitted values and against the bachelor's share.

Table 5. Ten most influential counties (Cook's distance)

	county	pop	poverty_rate	ba_share	median_age	rucc	cooks_d
2414	Oglala Lakota County, South Dakota	13165	57.6	11.8	28.2	9	0.022016
550	Kalawao County, Hawaii	66	22.7	22.4	39.5	3	0.015934
2034	Sioux County, North Dakota	3659	43.3	16.3	28.2	9	0.013491
2423	Todd County, South Dakota	9119	48.4	16.6	25.5	9	0.012699
102	Greenlee County, Arizona	9378	8.2	20.9	36.3	9	0.011857
1754	Eureka County, Nevada	1579	24.9	17.1	50.7	9	0.009963
2378	Corson County, South Dakota	3781	44.2	19.0	29.9	9	0.009351
2410	Mellette County, South Dakota	1888	41.7	19.1	37.5	9	0.008156
3121	Albany County, Wyoming	35570	20.9	55.0	28.6	4	0.008063
2398	Jackson County, South Dakota	2729	42.4	12.4	34.3	9	0.007735

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ba_share, all counties: −0.186
ba_share, without influential counties: −0.177  SE: 0.013
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